A. Test Cases and Risk Analysis

Kumparan QA Engineer Skill Test

Application: https://kumparan.com
Testing types: Manual, exploratory, functional, responsive UI, authorization, and negative testing
Target platforms: Desktop web and mobile web
Automation framework for subsequent tasks: Playwright with TypeScript

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1. Objective

The objective of this test activity is to evaluate the core functionality of the Kumparan website on desktop and mobile browsers. Testing focuses on user journeys, authorization, input validation, UI behavior, error handling, and responsive functionality.

The testing process aims to:

- Validate critical user journeys and business functionality.
- Identify defects and usability issues through exploratory testing.
- Prioritize test scenarios based on business impact and likelihood of failure.
- Establish test cases that can be automated using Playwright.
- Document assumptions, dependencies, test data, execution results, and exit criteria.

2. Scope

In Scope

- Homepage loading and navigation.
- Article access and content rendering.
- Search functionality and search result behavior.
- Article sharing controls.
- Like and comment interactions without submitting actual content.
- Report controls without submitting an actual report.
- Login and logout behavior.
- Invalid login handling.
- Desktop and mobile responsiveness.
- Error handling and unexpected page states.
- Input validation and boundary conditions.
- Basic accessibility and usability observations.

Out of Scope

- Actual submission of comments or article reports.
- Production data modification or destructive actions.
- Load, stress, and performance testing.
- Security penetration testing.
- Testing internal editorial or administrative features.
- Testing third-party authentication providers beyond the publicly available login flow.
- Testing features that require unavailable authorized accounts or special permissions.

3. Assumptions and Dependencies

Assumptions

1. The public Kumparan website is available during testing.
2. Desktop and mobile layouts may differ, provided core business functionality remains accessible.
3. Users must authenticate before accessing interactions that require an account.
4. Login with valid credentials should succeed, while invalid credentials should be rejected with a clear error message.
5. Sharing, liking, commenting, and reporting controls should be available according to the applicable page and user state.
6. Comments and reports must not actually be submitted during this assessment.
7. Search behavior, empty-state messaging, and maximum input length should follow the application's actual requirements. Where requirements are unavailable, findings will be documented as observations rather than confirmed defects.

Dependencies

- Stable internet connectivity.
- Access to the public Kumparan website.
- Desktop Chromium and mobile browser emulation.
- A valid test account and invalid test credentials, if provided and authorized.
- Playwright and TypeScript for subsequent automation.
- Availability of article content and search functionality.
- Third-party login providers, if authentication redirects outside Kumparan.

4. Risk Analysis and Prioritization

Priority is determined using potential business impact and likelihood of occurrence.

Priority	Risk level	Description
P1 – Critical	High	Core website access, authentication, or a major user journey is unavailable or behaves incorrectly.

P2 – High	Significant	Important functionality such as article access, search, sharing, or authorization is broken.
P3 – Medium	Moderate	Partial functionality, validation, responsive layout, or recoverable errors are affected.
P4 – Low	Low	Minor visual inconsistencies or usability issues with limited impact.

Priority represents the importance of the test scenario, not the severity of a defect that has already been confirmed.

5. Test Cases

A. Homepage and Navigation

ID	Test scenario	Steps	Expected result	Priority
TC-001	Homepage loads successfully	Open the homepage on desktop and mobile.	Homepage loads without a blocking error, and primary content is accessible.	P1
TC-002	Primary navigation works	Select available navigation links such as News, Business, or other categories.	Each link opens the corresponding destination.	P2
TC-003	Article link navigation	Select an article from a listing.	The correct article page opens and displays its title.	P2
TC-004	Browser back and forward navigation	Navigate from homepage to an article, then use Back and Forward.	The browser returns to the expected pages without an unrecoverable state.	P3
TC-005	Mobile navigation usability	Open the mobile homepage and inspect primary navigation controls.	Navigation remains visible or accessible through the mobile layout.	P2

B. Search Functionality

ID	Test scenario	Steps	Expected result	Priority
TC-006	Search with a valid keyword	Open search, enter data, and submit.	Search results or a relevant search state is displayed.	P2
TC-007	Search with no results	Search for a sufficiently unique keyword unlikely to exist.	An appropriate empty state is displayed without a page crash.	P3
TC-008	Search with an empty input	Open search and submit without entering a keyword.	The application handles the empty input according to its validation rules.	P3

TC-009	Search with whitespace	Enter leading/trailing spaces around a keyword.	The application handles whitespace consistently and does not crash.	P3
TC-010	Search with special characters	Enter characters such as @#$% as a keyword.	The application safely handles the input and displays results or an empty state.	P3
TC-011	Search with a long keyword	Enter a long string within and beyond any documented input limit.	The application handles boundary conditions without layout breakage or errors.	P3
TC-012	Search on mobile	Open the search destination using mobile navigation and search for a keyword.	Search is accessible and usable on the mobile layout.	P2

C. Article Content and Sharing

ID	Test scenario	Steps	Expected result	Priority
TC-013	Article content renders	Open a valid article URL.	The title, author information, and article content are displayed.	P1
TC-014	Article paragraphs are readable	Inspect the article body on desktop and mobile.	Text is readable, visible, and not significantly clipped or overlapped.	P2
TC-015	Share control availability	Inspect the article page for a share button or share icon.	A share control is available and identifiable.	P2
TC-016	Share interaction	Activate the share control without publishing or sending anything.	A share menu, share dialog, or supported sharing interface opens.	P2
TC-017	Share state can be dismissed	Open the share interface and close it using the available close action or Escape key.	The interface closes and the article remains usable.	P3
TC-018	Report control availability	Locate the “Laporkan tulisan” control on an article page.	The report control is visible and identifiable.	P2
TC-019	Report flow without submission	Open the report interface, inspect the available options, and exit without submitting.	The interface can be inspected and dismissed without creating a report.	P2

D. Like, Comment, and Authorization

ID	Test scenario	Steps	Expected result	Priority

TC-020	Like interaction as an unauthenticated user	Open an article and attempt to use the like control.	The application requests authentication or clearly prevents the protected action.	P2
TC-021	Comment interaction as an unauthenticated user	Open the comment interface and attempt to access the comment composer. Do not submit anything.	The application follows its authentication rules and clearly indicates if login is required.	P2
TC-022	Comment validation	If an authorized test account and comment composer are available, inspect empty and whitespace-only input without submitting.	Validation is appropriate and no comment is created.	P3
TC-023	Valid login	Use an authorized test account through the supported login flow.	Login succeeds and the authenticated state is displayed.	P1
TC-024	Invalid login	Use an approved invalid test account or invalid credentials.	Login is rejected with a clear error message and no authenticated session is created.	P1
TC-025	Logout	Log in with an authorized test account and select Logout.	The session ends and the unauthenticated state is restored.	P1
TC-026	Protected state after logout	Log out and attempt to access a protected interaction.	The application requires authentication again.	P2

E. Error Handling and Exploratory Testing

ID	Test scenario	Steps	Expected result	Priority
TC-027	Invalid article URL	Open a nonexistent article URL.	A suitable not-found or error page is shown without exposing internal errors.	P3
TC-028	Network interruption	Interrupt connectivity during a non-destructive page navigation, where safe and practical.	The application fails gracefully and provides a recoverable state.	P3
TC-029	Repeated navigation	Navigate between homepage, search, and article pages repeatedly.	The application remains stable without broken navigation.	P3

TC-030	Responsive layout inspection	Inspect key pages at desktop and mobile viewport sizes.	Content and controls remain usable without major overlapping or clipping.	P2
TC-031	Browser refresh	Refresh an article or search results page.	The page reloads successfully and preserves the expected URL and content state where applicable.	P3
TC-032	Exploratory interaction inspection	Explore visible controls, menus, article metadata, and page transitions.	Controls behave consistently with their labels and no unexpected destructive action occurs.	P3

6. Test Execution Results

Execution environment: Playwright with TypeScript; desktop Chromium and emulated mobile browser.

The following observations are based on the test runs and page output inspected during this assessment. These observations are not a substitute for executing every test case above.

Test area	Current observation	Status
Article access	The selected article title and article body were visible in the captured browser output.	Observed working
Article content	The article contained readable paragraphs and author information.	Observed working
Report control	The page text included “Laporkan tulisan”.	Observed working
Share control	The automated assertion for visible “Bagikan” or “Share” text failed. An icon or alternative share mechanism has not yet been ruled out.	Needs investigation
Like authorization	The automated test could not reliably confirm the unauthenticated interaction behavior.	Inconclusive
Desktop homepage	The page title and primary navigation were included in the automation checks.	Requires final run verification
Mobile homepage	Some desktop-oriented assertions, including “Buat Tulisan”, failed on mobile.	Test needs adjustment
Search	The search input locator and expected navigation state did not consistently match the mobile page.	Test needs investigation
Login and logout	Credential-based login depends on a supported authentication flow and authorized test credentials.	Dependency / not fully verified

Execution note: A failed assertion does not automatically establish a product defect. Some failures may be caused by an incorrect locator, different mobile layout, an unavailable control, or a test environment restriction. Each issue must be reproduced and classified before a final defect is confirmed.

7. Defect Reporting

The following issues are candidates for investigation. They should be promoted to confirmed defects only after the expected behavior is established and the behavior is reproduced.

Finding F-001: Share Control Not Identified by Text Locator

- **Area:** Article page / Sharing
- **Priority:** P2 investigation
- **Observation:** The Playwright assertion looking for Bagikan or Share in the page body failed.
- **Expected:** A share control should be available according to the test scenario.
- **Actual:** The expected text was not found in the rendered page text.
- **Impact:** The sharing feature could not be verified using the current locator.
- **Next steps:** Inspect accessible roles, button names, icons, and data-qa-id attributes. Verify the actual sharing interaction before confirming a defect.
- **Status:** Needs investigation.

Finding F-002: Mobile Navigation Assertion Is Desktop-Oriented

- **Area:** Mobile homepage / Navigation
- **Priority:** P3 test correction
- **Observation:** The mobile test expected the exact text Buat Tulisan, but the element was not found.
- **Expected:** Core navigation should remain usable on mobile.
- **Actual:** This specific desktop-oriented text assertion failed on mobile.
- **Impact:** The test cannot reliably evaluate mobile navigation using this assertion.
- **Next steps:** Identify the mobile navigation controls and verify their destination rather than requiring the same visible text as desktop.
- **Status:** Automation issue suspected; product defect not confirmed.

Finding F-003: Search Locator Does Not Match the Mobile Page

- **Area:** Mobile search
- **Priority:** P2 investigation
- **Observation:** The search input locator using the placeholder Cari di sini failed on the mobile page under test.
- **Expected:** Users should be able to reach and use the search feature on mobile.
- **Actual:** The expected input was not found at the locator's current page and state.

- **Impact:** The mobile search journey could not be verified.
- **Next steps:** Navigate through the actual mobile search control, inspect the destination page, and identify a stable locator.
- **Status:** Needs investigation.

8. Test Data Requirements

Data ID	Data	Purpose
TD-001	Existing public article URL	Validate article access and content rendering.
TD-002	Common keyword: data	Validate a normal search journey.
TD-003	Unique keyword	Validate empty search results.
TD-004	Special-character search string	Validate input handling.
TD-005	Long search string	Validate boundary behavior.
TD-006	Authorized test account	Validate login, logout, and protected actions.
TD-007	Approved invalid credentials	Validate rejected login and error messaging.
TD-008	Desktop and mobile viewport configurations	Validate responsive behavior.

Credentials must be supplied through a secure, approved mechanism and must not be committed to source control.

9. Test Environment

- **Website:** https://kumparan.com
- **Automation framework:** Playwright
- **Programming language:** TypeScript
- **Desktop project:** Desktop Chrome emulation
- **Mobile project:** iPhone 13 device emulation
- **API testing:** GraphQL endpoint handled separately in Task 4
- **Browser evidence:** Playwright HTML report, screenshots, videos, and traces when available

Device emulation validates responsive behavior but does not replace testing on physical mobile devices.

10. Execution Strategy and Test Isolation

- Separate desktop and mobile projects in Playwright.

- Use independent browser contexts to reduce shared cookies and session state.
- Use read-only public content wherever possible.
- Avoid submitting comments or reports.
- Use dedicated test accounts only when authorized.
- Keep test data independent and avoid relying on execution order.
- Use stable locators, prioritizing data-qa-id when available.
- Use role-based or label-based locators when suitable test IDs are unavailable.
- Capture screenshots and traces for failed tests.
- Investigate flaky tests before increasing retries or marking them as passed.

11. Exit Criteria

Testing can be considered complete when:

1. All planned in-scope test cases have an execution status.
2. P1 scenarios have been executed and their outcomes documented.
3. No unresolved critical defects remain without explicit acknowledgement.
4. Failed tests are classified as product defects, automation issues, environmental issues, or inconclusive.
5. Desktop and mobile results are reported separately.
6. Required evidence is attached for confirmed defects and significant failures.
7. Known limitations, skipped tests, and unavailable dependencies are documented.
8. The test report is reproducible by another team member.

12. Summary

This test plan covers the primary user journeys and risk areas of the Kumparan website across desktop and mobile environments. Initial browser observations confirmed that the selected article content was accessible and the report label was present. Search behavior, share control identification, mobile navigation, and authentication-dependent interactions require further verification.

The next step is to complete the remaining test executions, distinguish application defects from automation locator issues, and use the verified scenarios as the basis for the Playwright automation tasks.

Document status: Initial test plan and partial execution report.
Important: The execution status must be updated after the final test run; unexecuted scenarios must not be reported as passed.